



Koneru Lakshmaiah Education Foundation

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SECTION : DBSE&DBD Y25 Section-8

WEEK-3 QUESTION

H1

DBMS Lab: SQL Joins and Transactions

Part A: Create Tables

```
CREATE TABLE roles(role_id INT PRIMARY KEY, role_name VARCHAR(50));
CREATE TABLE users(user_id INT PRIMARY KEY, name VARCHAR(50), email VARCHAR(50), role_id INT);
CREATE TABLE departments(dept_id INT PRIMARY KEY, dept_name VARCHAR(50));
CREATE TABLE employees(emp_id INT PRIMARY KEY, emp_name VARCHAR(50), dept_id INT, manager_id INT, salary DECIMAL(10,2));
CREATE TABLE customers(customer_id INT PRIMARY KEY, customer_name VARCHAR(50), city VARCHAR(50));
CREATE TABLE orders(order_id INT PRIMARY KEY, customer_id INT, order_date DATE, total_amount DECIMAL(10,2));
CREATE TABLE accounts(account_no INT PRIMARY KEY, holder_name VARCHAR(50), balance DECIMAL(10,2));
```

Q1. Display employee names with their department names using INNER JOIN.

```
SELECT emp_name, dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;
```

Q2. Display all users with their assigned roles using LEFT JOIN.

```
SELECT name, email, role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id;
```

Q3. Display all roles even if no users are assigned using RIGHT JOIN.

```
SELECT name, role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
```

Q4. Simulate FULL OUTER JOIN in MySQL.

```
SELECT name, email, role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name, email, role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
```

Q5. Display employee and manager names using SELF JOIN.

```
SELECT e.emp_name Employee, m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;
```

Q6. Display OrderID, Customer Name, City and Total Amount.

```
SELECT o.order_id, customer_name, city, total_amount FROM customers c JOIN orders o ON c.customer_id=o.customer_id;
```

Q7. Find department-wise employee count, average salary and maximum salary.

```
SELECT dept_name, COUNT(*) AVG(salary), MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;
```

Q8. Display employees earning >50000 in CSE.

```
SELECT emp_name, salary, dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';
```

Q9. Display customer-wise total purchase in descending order.

```
SELECT customer_name, SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
```

Q10. Create Students, Courses, Faculty, Enrollments and display Student, Course, Faculty using joins.

```
SELECT student_name, course_name, faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
```

Part B: Transactions

Q11. Insert a new account and COMMIT.

```
START TRANSACTION; INSERT INTO accounts VALUES(104, 'Anil', 25000); COMMIT;
```

Q12. Insert a record and ROLLBACK.

```
START TRANSACTION; INSERT INTO accounts VALUES(105, 'Ravi', 10000); ROLLBACK;
```

Q13. Transfer ₹5000 from Account 101 to 102.

```
START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE account_no=101; UPDATE accounts SET balance=balance+5000 WHERE account_no=102; COMMIT;
```

Q14. Simulate failed transaction and rollback.

```
START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=101; UPDATE accounts SET balance=balance+3000 WHERE account_no=999; ROLLBACK;
```

Q15. Demonstrate SAVEPOINT.

```
START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE account_no=101; SAVEPOINT s1; UPDATE accounts SET balance=balance-1000 WHERE account_no=101; ROLLBACK TO s1; COMMIT;
```

DBMS Lab – SQL Joins and Transactions (Week 3)

Aim

To perform SQL JOIN operations and Transaction Control Language (TCL) commands using MySQL Programs (Q1–Q15)



Command Prompt - mysql -u



Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql>

mysql> ROLLBACK;

Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM accounts;

account_no	holder_name	balance
101	Ram	45000.00
102	Sita	35000.00
103	Hari	20000.00
104	Anil	25000.00

4 rows in set (0.00 sec)

mysql> START TRANSACTION;

Query OK, 0 rows affected (0.00 sec)

mysql>

mysql> UPDATE accounts

 -> SET balance = balance + 2000

 -> WHERE account_no = 101;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql>

mysql> SAVEPOINT s1;

Query OK, 0 rows affected (0.01 sec)

mysql>

mysql> UPDATE accounts

 -> SET balance = balance - 1000

 -> WHERE account_no = 101;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql>

mysql> ROLLBACK TO s1;

Query OK, 0 rows affected (0.00 sec)

mysql>

mysql> COMMIT; SELECT * FROM accounts;

Query OK, 0 rows affected (0.01 sec)

account_no	holder_name	balance
101	Ram	47000.00
102	Sita	35000.00
103	Hari	20000.00
104	Anil	25000.00



Command Prompt - mysql -u



```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
  -> SET balance = balance - 3000
  -> WHERE account_no = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE accounts
  -> SET balance = balance + 3000
  -> WHERE account_no = 999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Ram         | 45000.00 |
|          102 | Sita        | 35000.00 |
|          103 | Hari        | 20000.00 |
|          104 | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
  -> SET balance = balance + 2000
  -> WHERE account_no = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> UPDATE accounts
  -> SET balance = balance - 1000
  -> WHERE account_no = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;SELECT * FROM accounts;
Query OK, 0 rows affected (0.01 sec)
```

```
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Ram         | 47000.00 |
|          102 | Sita        | 35000.00 |
|          103 | Hari        | 20000.00 |
|          104 | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

Command Prompt - mysql -u

Query OK, 0 rows affected (0.00 sec)

account_no	holder_name	balance
101	Ram	45000.00
102	Sita	35000.00
103	Hari	20000.00
104	Anil	25000.00

4 rows in set (0.00 sec)

mysql> START TRANSACTION;

Query OK, 0 rows affected (0.00 sec)

mysql>

mysql> UPDATE accounts

-> SET balance = balance - 3000

-> WHERE account_no = 101;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql>

mysql> UPDATE accounts

-> SET balance = balance + 3000

-> WHERE account_no = 999;

Query OK, 0 rows affected (0.00 sec)

Rows matched: 0 Changed: 0 Warnings: 0

mysql>

mysql> ROLLBACK;

Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM accounts;

account_no	holder_name	balance
101	Ram	45000.00
102	Sita	35000.00
103	Hari	20000.00
104	Anil	25000.00

4 rows in set (0.00 sec)

mysql> |



Command Prompt - mysql -u X



```
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
101	Ram	50000.00
102	Sita	30000.00
103	Hari	20000.00
104	Anil	25000.00

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
101	Ram	50000.00
102	Sita	30000.00
103	Hari	20000.00
104	Anil	25000.00

```
4 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> UPDATE accounts
```

```
    -> SET balance = balance - 5000
```

```
    -> WHERE account_no = 101;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
```

```
mysql> UPDATE accounts
```

```
    -> SET balance = balance + 5000
```

```
    -> WHERE account_no = 102;
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
```

```
mysql> COMMIT;SELECT * FROM accounts;
```

```
Query OK, 0 rows affected (0.00 sec)
```

account_no	holder_name	balance
101	Ram	45000.00
102	Sita	35000.00
103	Hari	20000.00
104	Anil	25000.00

```
4 rows in set (0.00 sec)
```

```
mysql> Ss
```

```
Command Prompt - mysql -u × + ▾

    -> (2,'Dr. Reddy');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO courses VALUES
    -> (101,'DBMS',1),
    -> (102,'Java',2);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO students VALUES
    -> (1,'Kruthi'),
    -> (2,'Rahul'),
    -> (3,'Sneha');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO enrollments VALUES
    -> (1,101),
    -> (2,102),
    -> (3,101);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT student_name,course_name,faculty_name
    -> FROM students s
    -> JOIN enrollments e
    -> ON s.student_id=e.student_id
    -> JOIN courses c
    -> ON e.course_id=c.course_id
    -> JOIN faculty f
    -> ON c.faculty_id=f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Sneha        | DBMS        | Dr. Sharma   |
| Kruthi       | DBMS        | Dr. Sharma   |
| Rahul        | Java        | Dr. Reddy    |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO accounts
    -> VALUES(104,'Anil',25000);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101        | Ram        | 50000.00 |
| 102        | Sita       | 30000.00 |
| 103        | Hari       | 20000.00 |
| 104        | Anil       | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO accounts
    -> VALUES(105,'Ravi',10000);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101        | Ram        | 50000.00 |
| 102        | Sita       | 30000.00 |
| 103        | Hari       | 20000.00 |
| 104        | Anil       | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```



```
mysql> SELECT * FROM employees;
```

emp_id	emp_name	dept_id	manager_id	salary
1	Ramesh	1	NULL	80000.00
2	Sita	1	1	60000.00
3	Ravi	2	1	45000.00
4	Anil	3	1	55000.00

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT emp_name, dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id = d.dept_id;
```

emp_name	dept_name
Ramesh	CSE
Sita	CSE
Ravi	ECE
Anil	EEE

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	email	role_name
Kruthi	kruthi@gmail.com	Student
Rahul	rahul@gmail.com	Faculty
Anil	anil@gmail.com	Admin

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT name, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	role_name
Anil	Admin
Rahul	Faculty
Kruthi	Student

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE faculty(
->     faculty_id INT PRIMARY KEY,
->     faculty_name VARCHAR(50)
-> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE courses(
->     course_id INT PRIMARY KEY,
->     course_name VARCHAR(50),
->     faculty_id INT
-> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE students(
->     student_id INT PRIMARY KEY,
->     student_name VARCHAR(50)
-> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE enrollments(
->     student_id INT,
->     course_id INT
-> );
```

```
Command Prompt - mysql -u X + v
-> account_no INT PRIMARY KEY,
-> holder_name VARCHAR(50),
-> balance DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_dbms_week3 |
+-----+
| accounts              |
| customers             |
| departments           |
| employees             |
| orders                |
| roles                 |
| users                 |
+-----+
7 rows in set (0.02 sec)

mysql> INSERT INTO roles VALUES
-> (1,'Admin'),
-> (2,'Faculty'),
-> (3,'Student');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO users VALUES
-> (101,'Kruthi','kruthi@gmail.com',3),
-> (102,'Rahul','rahul@gmail.com',2),
-> (103,'Anil','anil@gmail.com',1);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO departments VALUES
-> (1,'CSE'),
-> (2,'ECE'),
-> (3,'EEE');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO employees VALUES
-> (1,'Ramesh',1,NULL,80000),
-> (2,'Sita',1,1,60000),
-> (3,'Ravi',2,1,45000),
-> (4,'Anil',3,1,55000);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO customers VALUES
-> (1,'Kruthi','Hyderabad'),
-> (2,'Rahul','Vijayawada'),
-> (3,'Sneha','Vizag');
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO orders VALUES
-> (101,1,'2026-08-01',2500),
-> (102,1,'2026-08-02',3500),
-> (103,2,'2026-08-03',1500);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> ^C
mysql>
mysql> INSERT INTO accounts VALUES
-> (101,'Ram',50000),
-> (102,'Sita',30000),
-> (103,'Hari',20000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM employees;
```

Command Prompt - mysql -u

Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kruthi>mysql -u root -p
'mysql' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\kruthi>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE dbms_week3;
Query OK, 1 row affected (0.03 sec)

mysql> USE dbms_week3;
Database changed
mysql> CREATE TABLE roles(
-> role_id INT PRIMARY KEY,
-> role_name VARCHAR(50)
->);
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE users(
-> user_id INT PRIMARY KEY,
-> name VARCHAR(50),
-> email VARCHAR(50),
-> role_id INT
->);
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE departments(
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(50)
->);
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE employees(
-> emp_id INT PRIMARY KEY,
-> emp_name VARCHAR(50),
-> dept_id INT,
-> manager_id INT,
-> salary DECIMAL(10,2)
->);
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE customers(
-> customer_id INT PRIMARY KEY,
-> customer_name VARCHAR(50),
-> city VARCHAR(50)
->);
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE orders(
-> order_id INT PRIMARY KEY,
-> customer_id INT,
-> order_date DATE,
-> total_amount DECIMAL(10,2)
->);
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE accounts(
-> account_no INT PRIMARY KEY,

